# Secure your WiFi with WPA2-Enterprise

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**Introduction:** The purpose of this lab is to setup a WPA/WPA2 Enterprise (TKIP/AES + EAP-TTLS/PEAP) wireless network using FreeRadius and OpenWRT on a Buffalo router. In addition, you will configure Linux/Windows/Android clients to connect to the network.

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## 1 Install Wpad, OpenSSL, and FreeRadius

All the required packages (including the OpenWRT firmware) are stored in the following directory: `http://csfaculty.tcu.edu/ma/cosc40033/tools/?dir=WPA2_Buffalo`. It is important that you

use the packages from this directory.

First, flash your Buffalo router with the openwrt-ar71xx-generic-wzr-hp-ag300h-squashfs-sysupgrade.bin if it is not already running OpenWRT. Remember to set the router's root password to be password using the Web GUI interface. Otherwise, you may be able to login using the SSH service. Please also make sure that sshd service is running as well.

## 1.1 Install Wpad

We start by installing WPA2 related packages. Log onto the router as the root user, and type:

```
opkg remove wpad-mini
```

Or, you can use the GUI as shown in Fig. 1.

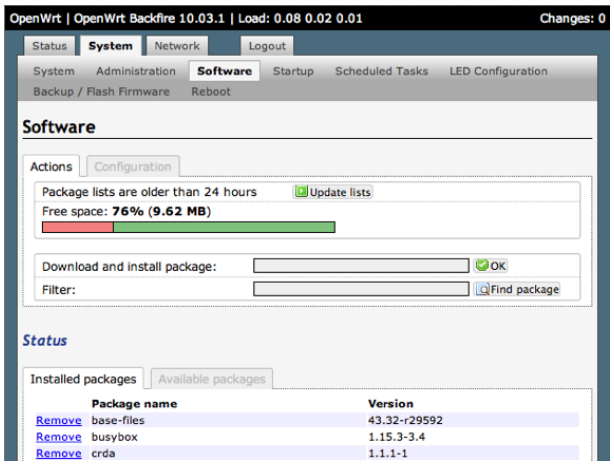


Figure 1: Package Management GUI.

Next, you need to upload all the packages inside the **Wpad** folder `http://csfaculty.tcu.edu/ma/cosc40033/tools/?dir=WPA2_Buffalo%2FPackages%2FWpad` onto the /tmp folder of the router. You can use the *scp* command to upload the files.

Once the upload is completed, type the following commands on your router:

```
cd /tmp/Wpad
opkg install  *.ipk
```

## 1.2 Install OpenSSL and FreeRadius

Similar to the installation of the wpad packages, you need to upload all the files of these two folders (OpenSSL and FreeRadius) to the /tmp folder of the router.
`http://csfaculty.tcu.edu/ma/cosc40033/tools/?dir=WPA2_Buffalo%2FPackages%2FOpenSSL`
`http://csfaculty.tcu.edu/ma/cosc40033/tools/?dir=WPA2_Buffalo%2FPackages%2FFreeRadius`

Once these files have been uploaded to the router, you can install them using the following command:

```
opkg install *.ipk
```

If no errors occur, we are done with the installation process!

### 1.3 Router Time Adjustment

Please set your router's time to the current time.

## 2 Configure a WPA2 Network

You need to setup a WPA2 enterprise enforced wireless network on your router. To do so, set the following parameters accordingly using either the GUI (as shown in Fig. 2) or command line (as shown in Fig. 3). Note you need to change the SSID according to your preference. You also can change the auth_secret if you wish.

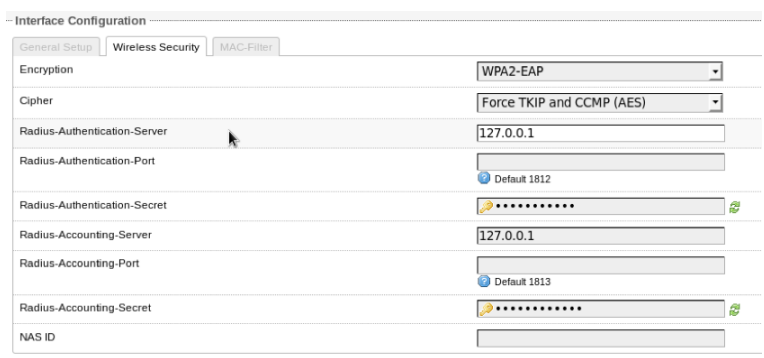


Figure 2: Web GUI to configure a WPA2 wireless network.

```
wireless.@wifi-iface[0]=wifi-iface
wireless.@wifi-iface[0].device=radio0
wireless.@wifi-iface[0].network=lan
wireless.@wifi-iface[0].mode=ap
wireless.@wifi-iface[0].ssid=40033-wpa2
wireless.@wifi-iface[0].encryption=wpa2+tkip+ccmp
wireless.@wifi-iface[0].auth_secret=myradiuskey
wireless.@wifi-iface[0].acct_secret=radiussecret
wireless.@wifi-iface[0].auth_server=127.0.0.1
wireless.@wifi-iface[0].acct_server=127.0.0.1
```

Figure 3: Command line to configure a WPA2 wireless network.

## 3 Configure FreeRadius

Configuring FreeRadius is the hardest part of this lab. It requires a lot of attention and patience. There are four files need to be configured for FreeRadius: *clients.conf, eap.conf, radiusd.conf* and

*users* (no suffix *.conf* for the *users* file). These four files are all located in the **/etc/freeradius2** directory on the router. Now, SSH login onto your router as root, and go to the /etc/freeradius2 directory on the router.

### 3.1 Edit clients.conf

The *clients.conf* controls what clients are allowed to query the Radius server. Thus, it needs to match the settings on the router such as the IP address and the radius authentication secret, which were setup in the previous section. It is recommended that you back up the original *clients.conf* file before any modifications. Now, you can directly work on the file to modify the following entries and keep other lines unchanged.

```
client localhost {
      ipaddr   = 127.0.0.1
      secret  = myradiuskey   #(You need to use the right secret)
```

### 3.2 Edit eap.conf

The *eap.conf* decribes what EAP methods are enabled and their corresponding settings. The peap protocol is enabled by default. You need to enable the ttls protocol (by removing the comments on ttl related lines) as well. For both protocols, please make sure that: the following variables are set to "yes":

```
copy_request_to_tunnel = yes
use_tunneled_reply = yes
```

### 3.3 Edit radiusd.conf

We need to make adjustments on the file to match our settings. Find the line consisting of interface = br-lan. Comment this line out.

```
#interface = br-lan
```

Comment out the entire second listen section:

```
# This second ``listen'' section is for listening on
# the accounting port, too.
#
# listen {
#     ipaddr = *
#     ipv6addr = ::
#     port = 0
#     type = acct
#     interface = br-lan
```

```
#     clients = per_socket_clients
# }
```

### 3.4 Edit users

In the end of the users file, add the following two lines:

```
testuser    Cleartext-Password := ``testpassword''
            Reply-Message = ``Hello, %{User-Name}''
```

### 3.5 Test Radius

Start the *radiusd* in the debug mode by issuing the following command:

```
radiusd -X
```

If there are errors in your configuration files, the debug mode will notify you. Otherwise, the radius daemon will go into sleep mode waiting authentication requests. It will output authentication related activities when the requests are received.

Next, we need to verify the settings of the radius server. Open another terminal on your computer and ssh to the router, and type the following command:

```
echo ``User-Name = testuser, User-Password = testpassword'' |
radclient -x 127.0.0.1 auth myradiuskey
```

If your radius settings are correct, you will see that the authentication request is accepted. On the other terminal (i.e., the one started radiusd -X), you will see outputs similar to in Fig. 4.
I upload the four files of my router to the following directory for your reference: `http://csfaculty.tcu.edu/ma/cosc40033/tools/?dir=WPA2_Buffalo%2FConfiguration_files`

#### 3.5.1 Provide a screenshot of your radiusd running and accepting incoming an authentication request from a wireless client.

## 4 Configure Linux Wireless Clients

To configure a Linux wireless client to connect to a WPA2 network, you may need to install/update *wpa_supplicant*. On your Linux, it should have been installed/updated. If not, type the following command as root: *yum install/update wpa_supplicant*.

Now, go to *Network Manager -> edit connections -> Wireless*, and add your wireless network with the necessary security parameters according to Fig. 5.

If your wireless client is connected, the connection information should look like Fig. **??**. If you choose PEAP instead of TTLS, it should work as well.

```
File  Edit  View  Terminal  Tabs  Help
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 127.0.0.1:2048, id=0, length=172
        User-Name = "testuser"
        NAS-IP-Address = 127.0.0.1
        Called-Station-Id = "00226b80ed1d"
        Calling-Station-Id = "001f3b6b17a9"
        NAS-Identifier = "00226b80ed1d"
        NAS-Port = 21
        Framed-MTU = 1400
        State = 0xd1778fc2a9fccd0d631718a948715e2e
        NAS-Port-Type = Wireless-802.11
        EAP-Message = 0x02090026190017030100 1b63095eb92041e0264cd2b9b9de9cc30722
67ef298db08c373e8103
        Message-Authenticator = 0xf023961eb613abec92ed4367c11bb2c2
  Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 9
  modcall[authorize]: module "mschap" returns noop for request 9
  rlm_eap: EAP packet type response id 9 length 38
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 9
    users: Matched entry testuser at line 73
  modcall[authorize]: module "files" returns ok for request 9
modcall: group authorize returns updated for request 9
  rad_check_password:  Found Auth-Type EAP
auth: type "EAP"
  Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 9
  rlm_eap: Request found, released from the list
  rlm_eap: EAP/peap
  rlm_eap: processing type peap
  rlm_eap_peap: Authenticate
  rlm_eap_tls: processing TLS
  eaptls_verify returned 7
  rlm_eap_tls: Done initial handshake
  eaptls_process returned 7
  rlm_eap_peap: EAPTLS_OK
  rlm_eap_peap: Session established.  Decoding tunneled attributes.
  rlm_eap_peap: Received EAP-TLV response.
```

Figure 4: Radius daemon output.

#### 4.0.1 Provide a screenshot of your established connection and the connection information.

#### 4.0.2 Under Linux, you do NOT need to specify TKIP or AES. Why? How do you know which encryption is chosen?

## 5 Configure Windows Wireless Clients (Optional)

Open the properties of your wireless network connection on your Windows desktop. Under the "Wireless Networks" tab, **add** your wireless network (example shown in Fig. 6).
Under the **association** tab, fill out the following information:

*Network name (SSID)*

*Select WPA for Network Authentication*

*Select TKIP for Data Encryption*

Then click on the **Authentication** tab at the top of this dialog-box. Select the options according to Fig. **??**. Basically, you need to choose PEAP as the EAP type. Under PEAP properties, uncheck the option of "Validate server certificate." In addition, choose EAP-MSCHAPv2 as the authentication method. Configure EAP-MSCHAPv2 to not use Windows logon name and password. Save all the settings and you are ready to connect to the network. We will create and deploy server certificates in the future sections.

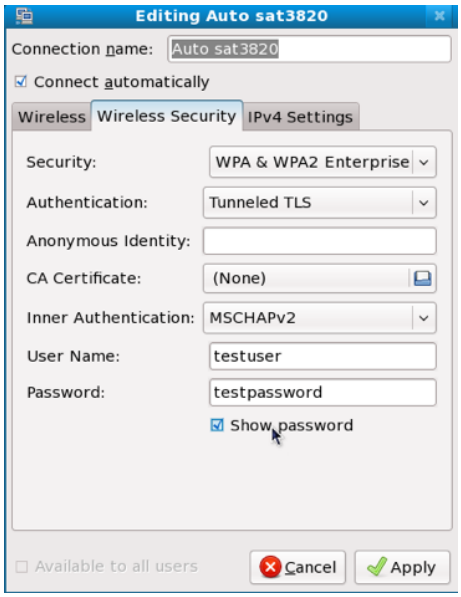


Figure 5: Wireless security parameters.

When you are connecting to your wireless network, there will be a pop-up window asking for credentials. Supply these credentials (i.e., user name and password) according to the actual settings. You do not need to fill the Logon domain. (Fig. 7)

#### 5.0.1 Provide a screenshot your connection status with the network properties displayed (WPA2+AES).

## 6 Running Radiusd as Daemon

If either Linux or Windows wireless client can connect to the wireless network successfully, you can make the radius a service on boot on the router. To achieve that, you need to type the following command on your AP:

```
cp -s /etc/init.d/radiusd /etc/rc.d/S70radiusd
```

You may also need to modify the */etc/init.d/radiusd* file. In the start function, change:

```
radiusd -i $IPADDR -p 1812,1813 $OPTIONS
to
radiusd -i 127.0.0.1 -p 1812,1813 $OPTIONS
```

Reboot the router to see whether your client can still connect.

## 7 Certificate Creation and Deployment

Previously, a wireless client cannot verify/accept the authenticity of an AP using the certificate automatically. Therefore, we need to generate a valid certificate on the AP so that the client can

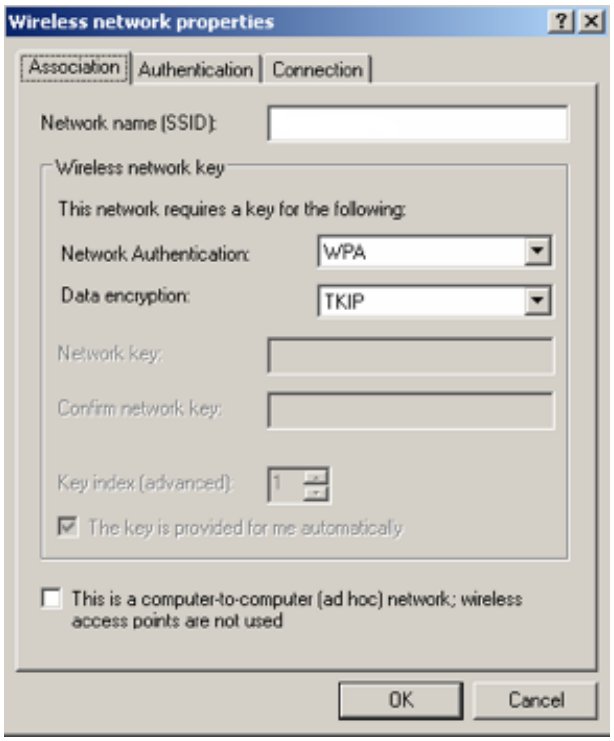


Figure 6: Wireless Network Properties.

automatically verify the AP's identity using the chain of trust within a CA (certificate authority). Setup and validate certificates may require a lot debugging efforts, though the theory behind it is relatively simple.

## 7.1 Become a CA and Create Certificates

**On your Linux desktop**, as root, go to the */etc/pki/tls/misc/* directory. There should be a file named *CA.pl*. If this file does not exist, you need to install additional openssl related packages on your Linux desktop. We will use this file to create the root certificate and other certificates.

**Create a new CA:**

```
./CA.pl -newca
    A certificate filename (or enter to create) <enter>

    Making CA certificate ...
    Using configuration from openssl.cnf
    Generating a 1024 bit RSA private key
    ............++++++
     .....................++++++
     writing new private key to `../../CA'
```

Figure 7: Credentials

```
Enter PEM pass phrase: <secret passphrase here>
 Choose your PEM, and write it down for future reference, mine is cosc40033test
Verifying password - Enter PEM pass phrase: <secret passphrase again>
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
-----
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:TX
Locality Name (eg, city) []:Fort Worth
Organization Name (eg, company) [Internet Widgits Pty Ltd]:TCU
Organizational Unit Name (eg, section) []:CITE
Common Name (eg, YOUR name) []: YourName #I put OpenWRT here
as an example common name. Later you will see this name again.
Email Address []:yourname@tcu.edu
```

You will be asked to input your PEM key again in the end (before the accomplish of the certificate creation)

This creates a new root certificate in the directory */etc/pki/CA*. The new root certificate file is named as *cacert.pem*.

**Next, generate a certificate request:**

```
./CA.pl -newreq

    Using configuration from openssl.cnf
```

```
Generating a 1024 bit RSA private key
..........++++++
.............++++++
writing new private key to `newreq.pem'
Enter PEM pass phrase: <another secret passphrase here> mine is wpa2cite
Verifying password - Enter PEM pass phrase: <another secret passphrase again>
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
-----
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:TX
Locality Name (eg, city) []:Fort Worth
Organization Name (eg, company) [Internet Widgits Pty Ltd]:TCU
Organizational Unit Name (eg, section) []:CITE
Common Name (eg, YOUR name) []:192.168.1.1, which is the router's IP address.
Our routers do not have a FQDN name yet.
Email Address []:yourname@tcu.edu

Please enter the following `extra' attributes
to be sent with your certificate request
A challenge password []:just hit <enter>
An optional company name []:just hit <enter>
Request (and private key) is in newreq.pem
```

The generated certificate request and private key in are stored in *newreq.pem*.

**Decoding the certificate request:**

```
openssl req -text -noout < newreq.pem
```

**Now, sign the certificate request:**

```
./CA.pl -sign
```

If you received no errors, the certificate has been signed by your root certificate. Under the */etc/pki/tls/misc/* directory, there will be the certificate, newcert.pem, and the private key - newreq.pem (encrypted) and newkey.pem (unencrypted). They are now ready to be used. You may wish to rename the files to more intuitive names. In addition, the root certificate file (named as *cacert.pem*) is located in the */etc/pki/CA* directory.

**Convert Root Certificate from PEM to DER format (for Windows clients):**

```
openssl x509 -in cacert.pem -inform PEM -out my-rootCA.der -outform DER
```

This will create a DER format certificate that can be easily installed on Windows.

### 7.2 Place the Root Certificate and the Server Certificate on Your Router

Upload three files: newkey.pem, newcert.pem, and cacert.pem onto your router. Place these three files under the */etc/freeradius2/certs* directory.

**Edit the *eap.conf* file in the following places:**

```
#private_key_password = whatever
private_key_password = wpa2cite #this is the private key for AP's certificate

#private_key_file = ${certdir}/server.pem
private_key_file = ${certdir}/newkey.pem

#certificate_file = ${certdir}/server.pem
certificate_file = ${certdir}/newcert.pem

#CA_file = ${cadir}/ca.pem
CA_file = ${cadir}/cacert.pem
```

Save the *eap.conf* file and reload the radius in the debug mode (radiusd -X). If there are no errors and the radiusd is accepting requests, your settings are correct.

## 8 Configure Certificate Authentication on a Linux Client

Under Linux, there is only one place you need to change for using the certificate: Add the corresponding *cacert.pem* into the properties of your wireless network (example shown in Fig. 8). Once finished, try connecting to your wireless network.

#### 8.0.1 Provide a screenshot of the configuration on your Linux desktop.

#### 8.0.2 Can you still capture/view the 4-way handshake? Why?

## 9 Configure Certificate Authentication on a Windows Client (Optional)

First, install the root certificate on the Windows client. Copy the file named *my-rootCA.der* to the Windows machine. Install the root certificate to establish ourselves as an authority.

1. Double-click on my-rootCA.der.

Figure 8: Add cacert.pem for Linux client.

2. In the "Certificate" property box, click Install Certificate.
3. In the Wizard, click Next.
4. Choose Place all certificates in the following store, and choose "Trusted Root Certification Authorities".
5. Click Next to finish.

Next, go to the properties tab of your wireless network (as shown in Fig. 9). Check the "validate server certificate" option and choose yourself created root CA (e.g., OpenWRT) as the trusted root CAs. Keep other settings unchanged.
Try to connect to your wireless network now. It should connect automatically if everything is setup correct.

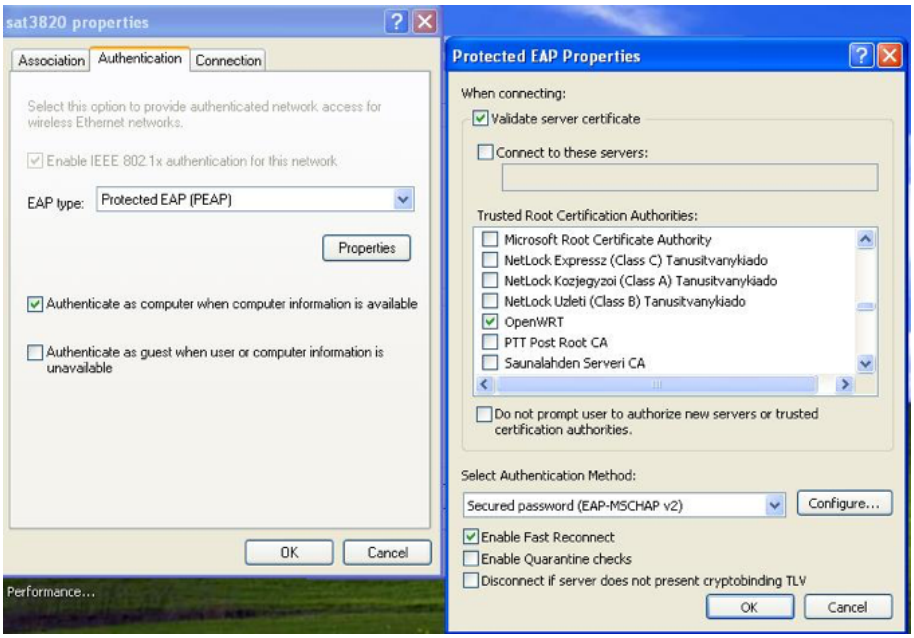


Figure 9: Windows Client Configuration.